Introduction to software testing

DINH THI MINH NGUYET

Learning outcomes

- Objectives of testing
- Type of software testing
- Software testing life cycle
- Test cases, defect logs and requirement traceability matrix
- Basic techniques to design test cases

What is software testing

It is the process used to identify the correctness, completeness and quality of developed computer software

It is the process of executing a program/application under positive and negative conditions by manual or automated means

Objectives

Uncover as many errors/bugs as possible in a given product

Demonstrate a given software product matching its requirement specifications

Validate the quality of a software testing using the minimum cost and efforts

Generate high quality test cases, perform effective tests, and issue correct and helpful problem reports

Levels of software testing

Unit testing: Validating that each software unit performs as expected. A unit is the smallest testable component of an application.

Integration testing: Ensuring that software components or functions operate together.

System Testing: System testing is types of testing where tester evaluates the whole system against the specified requirements.

Acceptance testing: Verifying whether the whole system works as intended.

Types of software testing

- Functional testing
  - Black box testing
  - White box testing
- Nonfunctional testing
  - Performance testing
  - Scalability testing
  - Usability testing
  - etc
- Maintenance testing
  - Regression testing
- Other types:
  - Smoke testing
  - Load testing
  - Stress testing

Software testing lifecycle

1. Requirement Analysis
2. Test Plan Creation
3. Environment setup
4. Test case design and execution
5. Defect Logging
6. Test Cycle Closure

Entry Criteria	Activities	Deliverable
For the planning of test plan requirement specification, application architecture document and well-defined acceptance criteria should be available.	Prepare the list of all requirements and queries, and get resolved from Technical Manager/Lead, System Architecture, Business Analyst and Client. Make a list of all types of tests (Performance, Functional and security) to be performed. Make a list of test environment details, which should contain all the necessary tools to execute test cases.	List of all the necessary tests for the testable requirements and Test environment details

1. Requirement Analysis

2. Test planning

- Preparation of test plan/strategy document for various types of testing
- Test toll selection
- Test effort estimation
- Resource planning and determining roles and responsibilities
- Training requirement

3. Environment setup

- Setup of the test environment
- Can be started along with Test Case Development.
- This is an essential part of the manual testing procedure as without environment testing is not possible.
- Environment setup requires a group of essential software and hardware to create a test environment.

4. Test case design and execution

- The testing team writes down the detailed test cases, also prepares the test data if required.
- The prepared test cases are reviewed by peer members of the team or Quality Assurance leader.
- Prepare RTM (Requirement Traceability Matrix used for tracking requirements. Each test case is mapped with the requirement specification. Backward & forward traceability can be done via RTM.

Defect Logging

- Evaluate the completion criteria of the software based on test coverage, quality, time consumption, cost, and critical business objectives.
- Defect logging analysis finds out defect distribution by categorizing in types and severity.

Test Cycle Closure

Evaluates the strategy of development, testing procedure, possible defects to use these practices in the future if there is a software with the same specification

Types of test document

- Test scenarios
- Test case
- Test plan
- Requirement traceability matrix(RTM)
- Test strategy
- Test data
- Bug report
- Test execution report

Test Case

The test case is defined as a group of conditions under which a tester determines whether a software application is working as per the customer's requirements or not

It is an in-details document that contains **all possible inputs (positive as well as negative) and the navigation steps**, which are used for the test execution process

When do we write a test case?

Test case template

Test cases design techniques

Error Guessing

Equivalence Partitioning

Boundary Value Analysis[BVA]

Error Guessing

The implementation of this technique depends on the experience of the tester or analyst having prior experience with similar application

The main purpose of this technique is to identify common errors at any level of testing by exercising the following tasks:

- Enter blank space into the text fields.
- Null pointer exception.
- Enter invalid parameters.
- Divide by zero.
- Use maximum limit of files to be uploaded.
- Check buttons without entering values.

Equivalence Partitioning

Input data is divided into ***partitions of valid and invalid values***, and it is mandatory that all partitions must exhibit the same behavior.

If a condition of one partition is true, then the condition of another equal partition must also be true, and if a condition of one partition is false, then the condition of another equal partition must also be false.

Example: Mobile number = 10 digits

INVALID	INVALID	VALID
DIGITS >=11	DIGITS<=9	DIGITS = 10

Boundary Value Analysis

It is used to test boundary values because the input values near the boundary have higher chances of error.

EX: Age is a variable of any function, and its minimum value is 18 and the maximum value is 30, both 18 and 30 will be considered as boundary values.

Agile software testing

Continuous integration and continuous delivery

Q&A